MILITARY SPECIFICATION

ANTENNA GROUP, OE-82( )/WSC-1(V)

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements for the fabrication, testing, and production of the OE-82( )/WSC-1(V) ultra-high frequency (UHF) satellite communication antenna group, hereinafter specified as the Antenna Group. The antennas are for use with the AN/WSC-3 Satellite Communication (SATCOM) Set and Line-of-Sight (LOS).

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-626	Brass, Leaded And Nonleaded, Rod, Shaped, Forgings, And Flat Products With Finished Edges (Bar And Strip)

MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment And Systems, Requirements For (Navy)
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-S-8660	Silicone Compound
MIL-P-15024	Plates, Tags, And Bands For Identification Of Equipment
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts, Packaging And Packing Of
MIL-P-24441	Paint, Epoxy-Polyamide, General Specification For
MIL-T-27730	Tape, Antiseize, Tetrafluoroethylene, With Dispenser
MIL-C-28838	Communication Sets, UHF-AN/WSC-3(V) Satellite And LOS Versions

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, Defense Standardization Program Branch, Department of the Navy, Washington, DC 20360, Attn: Elex 81111 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirements For Enclosures For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167	Mechanical Vibrations Of Shipboard Equipment (Type I- Environmental And Type II:- Internally Excited)
MIL-STD-118-114	Electrical Characteristics Of Digital Interface Circuits
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-681	Identification Coding And Application Of Hookup And Lead Wire
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1364	Standard General Purpose Electronic Test Equipment
MIL-STD-1399, Section 301	Interface Standard For Shipboard Systems, Ship Motion And Attitude
MIL-STD-1399, Section 302	Interface Standard For Shipboard Systems, Weather Environment

HANDBOOKS

MILITARY

MIL-HDBK-235, Part I	Electromagnetic (Radiated) Environment Considerations For Design And Procurement Of Electrical And Electronic Equipment Subsystems And Systems Part 1A
MIL-HDBK-235, Part II	Electromagnetic Radiation Environment From Friendly Or Own Force Emitters

2.1.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

HANDBOOKS

DEFENSE ATOMIC SUPPORT AGENCY

DASA - 2114-1	Electromagnetic Pulse (EMP) Hardening
DASA - 1420	Transient Radiation Effects (TREE) Hardening

PUBLICATION

NAVAL ELECTRONIC SYSTEMS COMMAND

NAVELEX PME116 letter 11019/SAL Ser S00860 Undated	Nuclear Environmental Statement

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(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Antenna group. The antenna group shall be capable of maintaining upper hemispherical coverage independent of motion and maneuver of the ship. The antenna group shall require no manual inputs or operational attendance for maintaining the antenna pointing to true satellite azimuth. Provision shall be made for manual azimuth positioning of the antenna indicators (manual azimuth and elevation) utilizing 115 Volt, 400 Hertz (Hz), single-phase ship power in lieu of the ship's gyro 400 Hz signals. This ship's power shall be used to energize the synchro repeater system while in the manual mode. The antenna group shall permit full duplex operation. The antenna group shall conform to the requirements of MIL-E-16400, except as hereinafter specified. The antenna group shall be compatible with the AN/WSC-3 as specified in MIL-C-28838 without danger of preamplifier burn-out and shall permit automatic switching between LOS and SATCOM.

3.1.1 Equipment composition. The antenna group shall consist of the following:

a. Two antenna units, AS-3018( )/WSC-1(V)
b. Two amplifier-filters (deck boxes), AM-6691( )/WSC-1(V)
c. One switching unit, SA-2000( )/WSC-1(V)
d. One antenna control, C-9597( )/WSC-1(V)

3.1.1.1 Antenna unit. The antenna unit shall conform to the requirements specified in 3.1.1.1.1 through 3.1.1.1.11.

3.1.1.1.1 Radiating element and reflector (radiating assembly). The radiating element and reflector of the antenna unit shall conform to the requirements specified in 3.1.1.1.1.1 through 3.1.1.1.1.12.

3.1.1.1.1.1 Power rating. Power handling rating shall be 800 watts (W) peak envelope power (PEP) and 400 W average, with no amplitude modulation of the radio frequency (RF) carrier.

3.1.1.1.1.2 Antenna coverage. Two antennas and associated pedestals normally will be used. The two antennas will be situated such that the satellite is always in view of at least one antenna. Hemispheric coverage from 2 degrees above horizon to zenith shall be required.

3.1.1.1.1.3 Frequency band. The frequency band shall be 240 to 318 megahertz (MHz).

3.1.1.1.1.4 Polarization. The polarization sense shall be right-hand circularly polarized for both transmit and receive.

3.1.1.1.1.5 Gain and sidelobes. The gain of the radiating assembly measured at the input connector shall be not less than 11 decibels (dB) over the frequency range 240 to 318 MHz and 12 dB at 305 MHz with respect to a circularly polarized isotropic radiator. The sidelobes shall be not less than 15 dB below the main beam maximum with the design goal of 18 dB.

3.1.1.1.1.6 Voltage standing wave ratio (VSWR). The VSWR measured at the input connector to the pedestal shall be not greater than 1.5:1 across the frequency range 240 to 290 MHz and 1.2:1 across the frequency range 290 to 318 MHz.

3.1.1.1.1.7 Axial ratio. The axial ratio over the frequency range specified in 3.1.1.1.1.5 shall be 3.0 dB or less within the 3 dB beamwidth.

3.1.1.1.1.8 Hybrids. Hybrids shall be provided as part of the antenna unit. Hybrid performance shall be such as to meet the requirements specified herein for the antenna unit.

3.1.1.1.1.9 Feedlines. Feedlines (coaxial cable) will be provided by the installation activity and are not required as part of the Contract End Item (CEI).

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3.1.1.1.1.10 Reflector. The back section of the reflector shall be no less than 0.080 inch thick aluminum. The seams shall be fully welded on the inner and outer perimeters.

3.1.1.1.1.11 Radiating element. The element base shall be anodized in accordance with MIL-A-8625, Type II, Class 1. The element base shall be sealed only if nickel acetate solution leaves no nickel plating on the radiating element. The element hybrid assembly shall be sealed with polyethylene terephthalate.

3.1.1.1.1.12 Mounting element. When radiating elements are mounted on the reflector assembly, the mounting surfaces shall be freshly coated with a non-metallic, pigmented epoxy paint, in accordance with MIL-P-24441.

3.1.1.1.2 Pedestal. The pedestal of the antenna unit shall conform to the requirements specified in 3.1.1.1.2.1 through 3.1.1.1.2.14.

3.1.1.1.2.1 Coverage. Each steerable pedestal shall be capable of providing full coverage of sky. Two mounts, if used, shall track identically and simultaneously in heading within system tolerances from the antenna control.

3.1.1.1.2.2 Axis. Two axis elevation over train pedestals shall be provided.

3.1.1.1.2.3 Accuracy. Each axis of the pedestal shall provide position data to an accuracy of one degree or better to the antenna control by means of synchro angle data pick-offs.

3.1.1.1.2.4 Travel. Each pedestal shall provide continuous unrestricted travel about the azimuth axis and from +2 degrees to +110 degrees in elevation, referred to the mounting plane.

3.1.1.1.2.5 Rotary joint. A rotary joint shall be provided in the azimuth axis. The rotary joint shall have a power handling capacity of 800 W PEP and 400 W average. The rotary joint shall have a wow of not more than 0.1 dB per 360 degrees rotation and an insertion loss of 1.0 dB or less across the frequency range of 225 to 400 MHz. Rotary joint drive pins shall be QQ-B-626 brass, alloy 260 or 268, hard tempered.

3.1.1.1.2.6 Slip rings. Slip rings in the train axis shall be provided for synchro information and power for the elevation drive motor.

3.1.1.1.2.7 Angular acceleration and velocity. The necessary angular acceleration and velocities required for the elevation and train axis drives shall be consistent with operating convenience, ship heading changes, environmental conditions, and pedestal weight. Angular velocity shall be not less than 3 degrees per second in azimuth and elevation.

3.1.1.1.2.8 Main beam axis. The pedestals for the antennas shall position the antenna main beam axis to within five degrees of the input commands from the antenna control.

3.1.1.1.2.9 Dead band. A dead band setting of seven degrees shall be incorporated into the design to alleviate excessive stress on the servosystem (azimuth and elevation drive motor).

3.1.1.1.2.10 Tracking. The equipments specified in 3.1.1 are not active tracking devices and do not use an RF signal originating from the satellite to detect the off-axis error and thereby make pointing corrections. Each pedestal shall respond to the command of an operator who inserts signals to the antenna control corresponding to true azimuth and elevation of the satellite. The azimuth of the antenna main beam axis shall be pointed in the correct direction after corrections have been made by the antenna control from the ship's heading reference.

3.1.1.1.2.11 Manual override. Provision shall be made for manually overriding the sector switch and selecting one of the antennas for operation. Provisions shall be made to automatically override the sector switch blockage of one antenna when the alternate antenna is disabled. This override control shall be mounted on the front of the antenna control and be clearly labeled.

3.1.1.1.2.12 Power rating. The power rating capacity shall be 400 W average and 800 W PEP with no amplitude modulation of the RF carrier.

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3.1.1.1.2.13 Telephone circuit. Circuitry for sound-power telephone communications between the pedestal and the antenna control shall be provided.

3.1.1.1.2.14 Sector switching. Sector switching circuitry shall be provided as an integral part of the pedestal for automatically switching from one to the other antenna. Antenna group design shall make maximum utilization of antenna pattern overlap to reduce switching occurrence. The switching zones shall be adjustable in increments of approximately seven degrees azimuth. An access opening shall be provided for use by installation personnel for sector switching adjustments during installation.

3.1.1.1.3 Weight. If the amplifier-filter is included within the pedestal, the combined weight shall not be greater than 425 pounds.

3.1.1.1.4 Antenna unit swept volume. The radius of the swept volume shall be not greater than 41 inches.

3.1.1.1.5 Mounting. Brackets, clamps, or other suitable means of attaching the antenna unit and pedestal to the recommended mounting surface shall be provided.

3.1.1.1.6 Connector. An EIA bulkhead connector, manufacturer number 26907, part numbers 42374; 32255, or equal shall be provided for the transmit feedline.

3.1.1.1.7 Counterbalance. Antenna unit-mounted modules and other components of the subsystem shall be used as counterbalances to the maximum extent possible to reduce dead weight and resultant increases in inertial loading. The smallest cantilever moment arms possible shall be used to minimize weight, inertia, and swept volume. Cable clamps shall be provided to reduce cantilever moment arms on connectors, as required, to withstand the shock requirement of MIL-E-16400 and the vibration requirement specified in 3.5.1.6.

3.1.1.1.8 Stops. Both electrical and mechanical stops shall be provided for the elevation axis. The mechanical stops shall operate without degradation of performance regardless of the frequency of use with the antenna being driven under full power. Means shall be provided for driving the elevation axis from the stop positions and to permit full drive power operation only in the out-of-stop direction.

3.1.1.1.9 Antenna unit blocked safety switch. Provision shall be made to disable the tracking mechanism and radio frequency feed of each pedestal for maintenance purposes. The other pedestal shall remain operational during a complete 360 degrees azimuth rotation when the selected pedestal is disabled.

3.1.1.1.10 Anodizing. After they are machined, all aluminum parts shall be anodized in accordance with MIL-A-8625, Type II, Class 1. The reflector assembly shall remain unsealed only if nickel acetate solution causes nickel plating to form over coating. The Type III, Class 1 sealing process shall be used for the upper and lower housings of the pedestal.

3.1.1.1.11 Silcon grease. Silicon grease as specified in MIL-S-8660 shall be applied to the threads of every screw used to mount major antenna sub-assemblies (for example hybrids, slip rings, and so forth).

3.1.1.2 Amplifier-filters (deck boxes). Two deck boxes shall be provided with each dual antenna group. Each deck box shall include a preamplifier and diplexer. If practicable, the amplifier-filters shall be incorporated within the antenna pedestal, thereby eliminating the need for deck boxes. The amplifier-filters shall conform to the requirements specified in 3.1.1.2.1 through 3.1.1.2.2.7.

3.1.1.2.1 Preamplifier. The preamplifier shall conform to the requirements specified in 3.1.1.2.1.1 through 3.1.1.2.1.8

3.1.1.2.1.1 Preamplifier frequency range. The frequency range of the preamplifier shall be 248.5 to 270.5 MHz.

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3.1.1.2.1.2 Preamplifier band pass characteristic. One dB band-width shall be not less than 22 MHz centered about 259.5 MHz.

3.1.1.2.1.3 Preamplifier gain. The gain shall be 26.2 dB ±2.5 dB at center frequency at initial operation and after 1,000 hours of operation. Gain over the pass-band shall be within one dB of the gain at center frequency.

3.1.1.2.1.4 Preamplifier noise figure. The noise figure over the pass-band shall not exceed 3.5 dB both at initial operation and after 1,000 hours of operation.

3.1.1.2.1.5 Preamplifier intermodulation. When connected to a 50 ohm (Ω) resistive load with a VSWR not exceeding 2.0:1, third-order intermodulation products ($2f_2-f_1$), ($2f_1-f_2$) shall be not greater than -85 decibel referred to one milliwatt (dBm) measured at the output of the preamplifier when two -40 dBm input signals (f_1 and f_2) located in the operating frequency range are amplified.

3.1.1.2.1.6. Preamplifier input impedance. Input impedance shall be nominally 50 Ω resistive with a VSWR not to exceed 2.0:1.

3.1.1.2.1.7 Preamplifier output impedance. Output impedance shall be nominally 50 Ω resistive with a VSWR not to exceed 1.5:1.

3.1.1.2.1.8 Preamplifier burnout protection. Burnout protection circuitry shall be located to protect the preamplifier and the remainder of the receiver chain. The frequency range of the burnout protection shall be from 225 MHz to 275 MHz. The burnout protection shall possess the following properties:

a. Strong interfering signals up to +38 dBm shall be limited to a level which shall not cause the receiver performance (signal to noise ratio) to be degraded by more than 0.2 dB.
b. Insertion loss at normal signal level shall be not greater than 0.4 dB.
c. Insertion loss at high signal levels shall be great enough to prevent damage to the receiver modules.
d. Input and output impedances of the burnout protectors shall be a nominal 50 Ω with a VSWR not to exceed 1.5:1.

3.1.1.2.2 Diplexer. The diplexer shall conform to the requirements specified in 3.1.1.2.2.1 through 3.1.1.2.2.7.

3.1.1.2.2.1 Diplexer transmit ripple. The ripple shall be not greater than 0.50 dB over the transmit band of 292.2 to 311.6 MHz.

3.1.1.2.2.2 Diplexer transmit insertion loss. The insertion loss shall not exceed 0.7 dB over the 19.4 MHz band specified in 3.1.1.2.2.1.

3.1.1.2.2.3 Diplexer receiver ripple. The ripple shall be not greater than ±0.50 dB over frequency band 248.5 to 270.1 MHz.

3.1.1.2.2.4 Diplexer receive channel insertion loss. The insertion loss shall be not greater than 1.0 dB over frequency band 248.5 to 270.1 MHz.

3.1.1.2.2.5 Diplexer transmit/receiver channel isolation. The receive channel shall provide at least 60 dB attenuation for all frequencies from direct current (DC) to 205 MHz. The isolation from the diplexer common port to the receiver port shall be not less than 95 dB from 205 to 225 MHz. This filter attenuation shall be as high as practical from 225 to 248.5 MHz, but consistent with the insertion loss of the receive pass-band. Similarly, from 270.1 to 285 MHz the filter attenuation shall be as high as practical, but consistent with the insertion loss of the receive band. From 285 MHz to 400 MHz, the attenuation shall be not less than 60 dB except for the transmit pass-band (292.2 to 311.6 MHz) where it shall be 100 dB. An attenuation of not less than 60 dB shall be required from 500 MHz to one gigahertz (GHz) and not less than 40 dB between one GHz and 10 GHz, except between two GHz and three GHz where the attenuation shall be not less than

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20 dB. Coupling from the transmit to receive channels shall be attenuated not less than 100 dB in the transmit band and shall be not less than 90 dB for any frequency in the receive band. The transmit channel shall provide not less than 30 dB attenuation for the third harmonic of the frequencies of the transmit band.

3.1.1.2.2.6 Diplexer VSWR. The VSWR of the receive ports shall be less than 1.5:1 in their frequency bands for a 50 Ω impedance line with remaining ports matched. The VSWR of the transmit ports shall be less than 1.2:1 in their frequency bands for a 50 Ω impedance line with remaining ports matched.

3.1.1.2.2.7 Diplexer power capability. The diplexer shall, as a minimum, be capable of operating with an average transmit power of 400 W and a PEP of 800 W.

3.1.1.3 Switching unit. The switching unit shall conform to the requirements specified in 3.1.1.3.1 through 3.1.1.3.7.

3.1.1.3.1 Transmit-transfer relay (antenna relay). The transmit-transfer relay shall be an integral part of the switching unit and shall be used for automatically switching from one antenna to the other. The transmit-transfer relay shall conform to the requirements specified in 3.1.1.3.1.1 through 3.1.1.3.1.6.

3.1.1.3.1.1 Function. The transmit transfer relay shall connect the desired antenna subsystem radio. It shall perform make-before-break function in order that an open circuit is never presented to the transmitter.

3.1.1.3.1.2 Switching time. Double termination of the transmission line shall not exist for longer than 10 milliseconds (ms) during the make-before-break interval provided by the antenna control.

3.1.1.3.1.3 Power. The contacts shall be capable of handling 800 W PEP and 400 W average.

3.1.1.3.1.4 VSWR. The VSWR of all ports shall be 1.15:1 with the ports properly terminated.

3.1.1.3.1.5 Insertion loss. The insertion loss shall be not greater than 0.2 dB.

3.1.1.3.1.6 Isolation between ports. Isolation from port to port shall be not less than 50 dB.

3.1.1.3.2 Receiver-transfer relay (antenna relay). The receiver-transfer relay shall be an integral part of the switching unit and shall be used for automatically switching from one antenna to the other. The receiver-transfer relay shall conform to the requirements specified in 3.1.1.3.2.1 through 3.1.1.3.2.4.

3.1.1.3.2.1 Switching time. The switching time shall be not greater than 25 ms.

3.1.1.3.2.2 Insertion loss. The insertion loss shall be not greater than 0.2 dB.

3.1.1.3.2.3 VSWR. VSWR shall be not greater than 1.2:1.

3.1.1.3.2.4 Power. The contacts shall be capable of handling 800 W PEP and 400 W average.

3.1.1.3.3 SATCOM/LOS relay. The SATCOM/LOS relay shall be an integral part of the switching unit to switch the AN/WSC-3 between SATCOM/LOS antenna relays. The SATCOM/LOS relay shall conform to the requirements specified in 3.1.1.3.3.1 through 3.1.1.3.3.5.

3.1.1.3.3.1 Isolation between ports. The isolation between ports shall be not less than 60 dB.

3.1.1.3.3.2 Switching time. The switching time shall be not greater than 25 ms.

3.1.1.3.3.3 Insertion loss. The insertion loss shall be not greater than 0.2 dB.

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3.1.1.3.3.4 VSWR. VSWR shall be not greater than 1.2:1.

3.1.1.3.3.5 Power. The contacts shall be capable of handling 800 W PEP and 400 W average.

3.1.1.3.4 LOS transmit/receive relay. The LOS transmit/receive relay shall be an integral part of the switching unit to switch the LOS antenna between receive and transmit and shall be as specified in 3.11.3.4.1 through 3.1.1.3.4.7.

3.11.3.4.1 Isolation between ports. The isolation between ports shall be not less than 60 dB.

3.1.1.3.4.2 Switching time. The switching time shall be not greater than 500 microseconds (μ s).

3.1.1.3.4.3 Insertion loss. The insertion loss shall be not greater than 0.2 dB.

3.1.1.3.4.4 VSWR. VSWR shall be not greater than 1.2:1.

3.1.1.3.4.5 Power. The contacts shall be capable of handling 800 W PEP and 400 W average.

3.1.1.3.4.6 Operating voltage. The LOS transmit/receive relay shall operate from existing 28 VDC power supply or 115 VAC or both.

3.1.1.3.4.7' Relay control. The relay shall operate from ground command and MIL-STD-188-114 (+6 VDC enables relay, -6 VDC disables relay).

3.1.1.3.5 28 Volt DC power supply. A 28 $\pm$3 volt DC power supply shall be provided as an integral part of the switching unit to power coaxial relays and the preamplifier.

3.1.1.3.6 Stepdown transformer. A stepdown transformer shall be provided as an integral part of the switching unit to accept shipboard primary power and transform to the required voltage to power the respective components. The stepdown transformer shall conform to the requirements specified in 3.1.1.3.6.1 and 3.1.1.3.6.2.

3.1.1.3.6.1 Input voltage. The input voltage shall be 440 volts, three phase delta, 50 to 60 Hz.

3.1.1.3.6.2 Output voltage. The output voltage shall be 208 volts, three phase wye, 50 to 60 Hz.

3.1.1.3.7 Weight. The weight of the switching unit shall not exceed 107 pounds.

3.1.1.4 Antenna control. An antenna control, configured for 19 inch cabinet installation, shall be provided for use by the operator for control of the antenna group. The antenna control shall conform to the requirements specified in 3.1.1.4.1 through 3.1.1.4.1.3.

3.1.1.4.1 Antenna control function. The antenna control shall include automatic antenna sector switching function, mount control necessary to accomplish the subsystem function, and manual azimuth slewing control. It shall control either or both pedestals. The antenna control shall process input commands of true elevation and azimuth, inserted by the operator, and ship heading from the ship's gyro compass, and convert them to commands for antenna pointing. The antenna control shall provide for slewing manually in azimuth both antennas independent of the ship's gyro and independent of each other. Relative positioning shall be provided by utilizing the ship's 115 volt, 400 Hz, single-phase power in lieu of ship's gyro signals.

3.1.1.4.1.1 Controls and indicators. Controls and indicators shall provide all controls, indications, and functions as indicated in TABLE I and FIGURE 1.

3.1.1.4.1.2 Telephone circuit. The telephone circuit from the pedestal shall be terminated at the antenna control.

3.1.1.4.1.3 Weight. The weight of the antenna control shall not exceed 18 pounds.

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TABLE I. Antenna control C-9597( )/WSC-1(V), controls and indicators.

Control or indicator marking	Type	Function
POWER ON-OFF	Toggle switch	Controls 28 volts direct current (Vdc) power and 115 volts alternating current (Vac), 400 Hz power to antenna control and pedestal.
POWER ON	Indicator (green)	When lit, indicates DC power is applied to antenna control and pedestal(s) and 115 Vac, 400 Hz ship power and Own Ships Course information are energized provided the two READY lamps are also lit.
ELEVATION	Control knob	Selects commanded antenna position in the elevation axis between 2 degrees (horizontal) and 110 degrees (beyond zenith).
ANTENNA SELECT IN USE (NO. 1)	Indicator (white)	When lit, indicates that antenna No. 1 is connected to the radio set group, specified in MIL-C-28838.
OVRD RF BLK (NO. 1) DUAL (NO. 2) OVRD RF BLK	Rotary switch	Permits selection of either or both antennas (in a two-antenna installation), or the connected antenna in a one-antenna installation, with the radio set group. In DUAL position, antenna not blocked is automatically selected for use. If neither is blocked, No. 1 antenna is preferred unless it had previously been blocked. If both antennas are blocked, the ANTENNA BLOCKED lamp is lit (see ANTENNA BLOCKED). If No. 1 (or No. 2) position is selected, antenna No. 1 (or No. 2) is manually selected and power is removed from the control circuits of the alternate antenna. If OVRD BLK position No. 1 (or No. 2) is selected, contacts of the ANTENNA SELECT switch bridge the sector switch contacts in the antenna pedestal to maintain the selected antenna operative even though the blocked sector is entered.
IN USE (NO. 2)	Indicator (white)	When lit, indicates that antenna No. 2 is connected to the radio set group.
AZIMUTH	Control knob	Selects commanded antenna position in the azimuth axis relative to true north (when MODE switch is at AUTO), or to ship heading (when MODE switch is at MANUAL).

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TABLE I. Antenna control C-9597( )/WSC-1(V), controls and indicators. (Continued)

Control or indicator marking	Type	Function
COAX PRESSURE	Indicator	When lit, indicates that the pressure in the dry air system is below 1.5 pounds per square inch gauge (psig).
ANTENNA BLOCKED	Indicator (red)	When lit, indicates that both antennas of a two-antenna installation are blocked or that the single antenna of a one-antenna installation is blocked. (If one antenna of a two-antenna installation is shut down for maintenance and ANTENNA SELECT switch is at OVRD RF BLK position, the ANTENNA BLOCKED lamp will not light when the selected antenna is in the blocked sector.)
PHONE	Telephone jack	Provides access to sound-powered intercom circuit with both antenna pedestals.
AZIMUTH ERROR	Indicator (red)	When lit, indicates a discrepency between the AZIMITH command angle and the selected antenna position. Lamp is out after antenna moves to the commanded azimuth position.
MODE READY, AUTO	Indicator	When lit, indicates that Own Ships Course information is present at the input of the antenna control.
MANUAL, AUTO	Toggle switch	In AUTO position, Own Ships Course information is used as a reference for azimuth command signals. In MANUAL position, a ship source of 400 Hz voltage without the gyro information superimposed is used to provide a reference based only on a relative heading of the vessel.
READY, MANUAL	Indicator	When lit, indicates that the ship source of 400 Hz voltage without Own Ships Course information upon it is available at the input to the antenna control.
ELEVATION ERROR	Indicator (red)	When lit, indicates a discrepancy between the ELEVATION command angle and the selected antenna position. Lamp is out after antenna moves to the commanded elevation position.
2A	Fuse holder	Contains 2-ampere fuse that protects the 28 Vdc input circuits of the antenna control.

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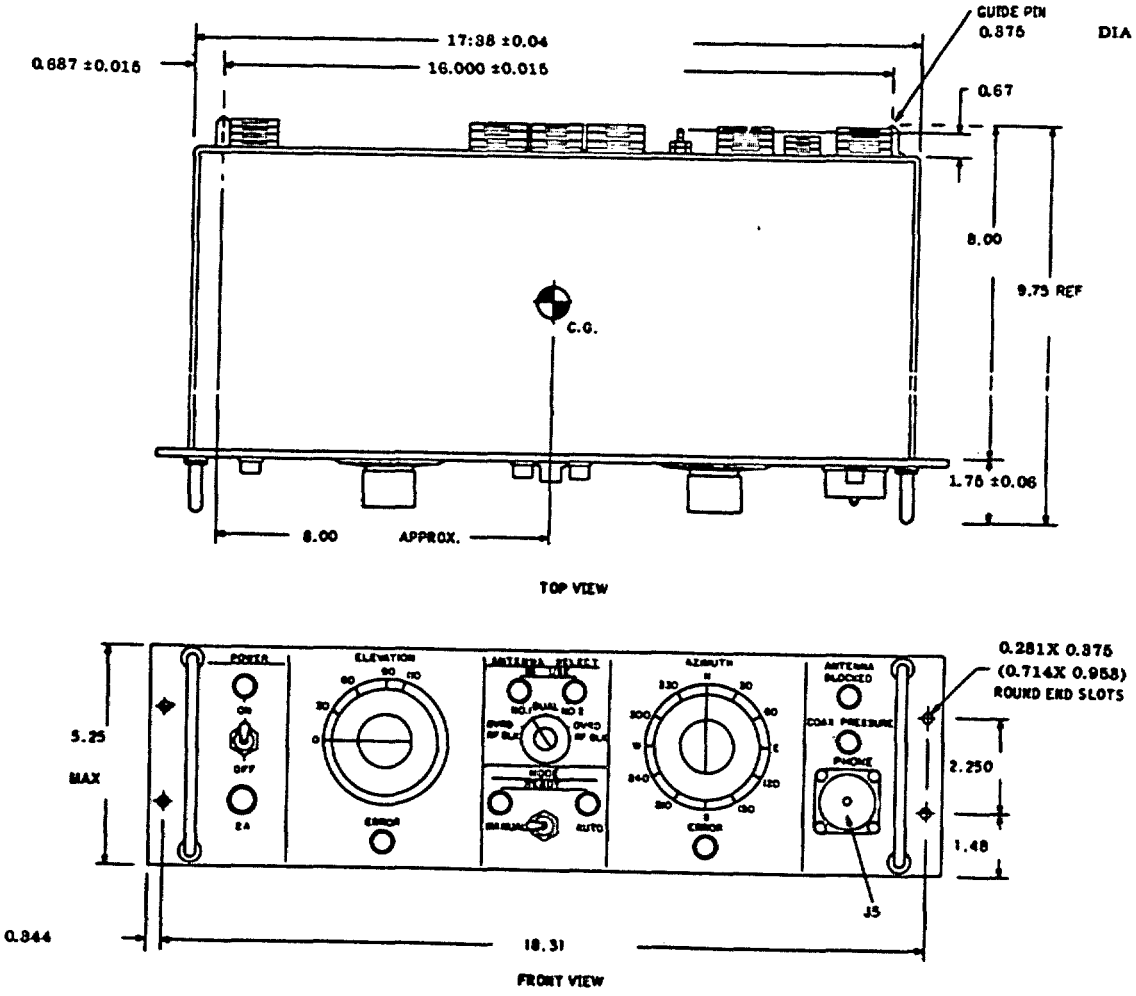


FIGURE 1. Antenna control, C-9597( )/WSC-1(V) panel layout and dimensions.

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3.2 First article. When specified, (see 6.2) a sample shall be subjected to first article inspection (see 4.3 and 6.4).

3.3 Reliability. The antenna group shall conform to the reliability requirements specified in 3.3.1 and 3.3.2.

3.3.1 Quantitative reliability. The upper mean-time-beteen-failure (MTBF) of the antenna group (θ_0 as defined by MIL-STD-781) shall be 1000 hours and the lower MTBF (θ_1 as defined by MIL-STD-781) shall be 500 hours.

3.4 Quantitative maintainability. Ease of maintenance, trouble-shooting and repair shall be a primary design consideration. Panels and doors which must be opened for maintenance or inspection more often than monthly shall have a maximum time to open of five seconds. Maintenance and repair shall be accomplished to the maximum extent practicable when the system is operational. The antenna group shall have a mean corrective maintenance time (M_{ct}) of 2.5 hours and a maximum corrective maintenance time (M_{max}) of 7.5 hours at a 95 percent confidence level when repair is accomplished by replacement of modules and chassis mounted components (this includes electronic and electro-mechanical parts, but excludes gears, radiating assemblies and so forth). The M_{ct} includes localization, isolation, disassembly, interchange, reassembly, alignment and checkout of all maintenance tasks. This excludes all preventive maintenance associated with the antenna group. Preventive maintenance shall be such that maximum outage time for preventive maintenance shall not exceed 45 minutes over a 180 day period.

3.5 Environmental requirements. The above deck and below deck equipments shall conform to the requirements specified in 3.5.1 through 3.5.2.

3.5.1 Above-deck environment. The above-deck equipments (antennas, pedestals, and deck boxes) shall conform to the requirements specified in 3.5.1.1 through 3.5.1.15.

3.5.1.1 Temperature range. The above-deck equipments shall conform to the Class 2 requirements of MIL-E-16400.

3.5.1.2 Salt fog. The above-deck equipments shall withstand the testing specified in 4.5.4 herein.

3.5.1.3 Altitude. The above-deck equipments shall operate normally at pressure equivalent to 5000 feet above sea level. The above-deck equipments shall survive, non-operating, pressure equivalent to 50,000 feet above sea level.

3.5.1.4 Wind. The above-deck equipments shall operate normally with wind loads of 75 knots. The above-deck equipments shall survive wind loads, non-operating, of 125 knots.

3.5.1.5 Ice loading. The above-deck equipments shall operate normally with ice loading of 4.5 pounds per square foot.

3.5.1.6 Sea state. The above-deck equipments shall operate normally under sea state five as specified in MIL-STD-1399 Sections 301 and 302. The above-deck equipments shall survive, non-operating, sea state nine.

3.5.1.7 Shock. The above-deck equipments shall conform to the shock requirements of MIL-S-901 for Grade A.

3.5.1.8 Vibration. The above-deck equipments shall conform to the vibration requirement of MIL-E-16400. Maximum frequency shall be 33 Hz as specified in MIL-STD-167.

3.5.1.9 Radio frequency interference. The above-deck equipments shall be self-compatible and meet the following requirements of MIL-STD-461: CE01, CE03, RE01, RE02, RS03 (using MIL-HBK-235 as guidelines), CS01, and CS06 (at 250 volts spike).

3.5.1.10 Enclosure. The above-deck equipments shall be spraytight in accordance with MIL-STD-108.

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3.5.1.11. Humidity. The above-deck equipments shall conform to the humidity requirements of MIL-E-16400.

3.5.1.12 Inclination. The inclination (degree) requirements shall be equivalent to the sea state conditions as specified in 3.5.1.6.

3.5.1.13 Nuclear effects electromagnetic pulse (EMP) hardening. The electromagnetic pulse threat levels shall be as specified in NAVELEX PME116 Ser S00860 letter. The above-deck equipments shall survive the specified EMP threat level environment without degradation of any requirements of the equipment, as specified herein, or require any repair actions necessary to restore full performance capabilities of the equipment. Uninterrupted operation of the set during the EMP is not required. The engineering practices in DSA 2114-1 pertinent to the hardening to EMP shall be utilized.

3.5.1.14 Nuclear transient radiation effects (TREE) hardening. The above-deck equipment shall survive the transient radiation effects on electronics (TREE) exposure levels specified in NAVELEX PME116 Ser S00860 letter. The above-deck equipments are not required to operate while subjected to the TREE environment. The TREE hardening techniques shall comply with DASA 1420. Exposure, without the benefit of shielding from nuclear radiation other than that employed in the design and fabrication of the antenna group, shall not result in causing either direct permanent damage effects or secondary damage which results in any degradation of performance of the above-deck equipments or in requiring any repair actions necessary to restore full performance capabilities of the above-deck equipments as specified herein.

3.5.1.15 Nuclear air blast. The above-deck equipments shall comply with the nuclear air blast requirements of MIL-E-16400, with the following values P = 3 pounds per square inch (psi), q = 0.2 psi, t+ = 3.5 (seconds) sec.

3.5.2 Below-deck environment. The below-deck equipments (switching unit and antenna control) shall conform to the environmental requirements of 3.5.1 except that the wind, ice loading and nuclear air blast requirements shall not apply.

3.6 Parts, materials and processes. The antenna group shall conform to the requirements specified in 3.6.1 through 3.6.9.

3.6.1 Selection of parts. All parts and materials shall be selected and applied in accordance with MIL-E-16400. Parts shall be derated in usage to provide a minimum safety factor as specified in the Appendix. Parts shall be rated at 125°C minimum, or higher if required by circuit application. Plastic encapsulated micro-circuits shall not be used.

3.6.2 Parts control. The parts to be incorporated in the antenna group shall be controlled in accordance with MIL-STD-965, Procedure I.

3.6.3 Metal parts. Metal parts used in fabrication of this equipment shall be of the corrosion-resistant type described in MIL-E-16400, or shall be plated to provide resistance to corrosion.

3.6.4 Moisture and fungus resistance. Moisture and fungus resistance shall be in accordance with Requirement 4 of MIL-STD-454.

3.6.5 Interchangeability, replaceability and repairability. Circuits shall be modularized so that similar functions shall use interchangeable modules. Replaced modules shall not require adjustment when integrated into the configuration. Access doors, if required, shall be nonstructural and installed with the minimum number of fasteners.

3.6.6 Finish. The exterior of the antenna units and the deck boxes shall be painted gray in accordance with MIL-P-24441. The exterior of the antenna control and switching unit shall be painted in accordance with MIL-E-15090, Class 2 (semi gloss), Type III (baking) enamel.

3.6.7 Attachment fittings. Attachment fittings or lifting eyes shall be provided for handling components.

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3.6.8 Access. Connectors shall be located and spaced to allow ease of handling by maintenance personnel wearing gloves. Easy access to all components for maintenance shall be provided to the maximum extent practicable.

3.6.9 Antiseize tape. Antiseize tape as specified in MIL-T-27730 shall be used for all pipe-threaded hardware into aluminum housings.

3.7 Design and construction. The antenna group shall conform to the requirements specified in 3.7.1 through 3.7.4.

3.7.1 Accessibility. The controls shall be readily accessible, simple to operate, and can be maintained by a two-person crew over any 12 hour period. Input and output terminals for the antenna control shall be provided at the rear of the chassis; however, maintenance shall be provided from the front only of the cabinet assembly when the antenna control drawer is withdrawn.

3.7.2 Maintenance. The maintenance concept for repair of failures in the antenna group shall be organizational replacement of plug-in type modules, PC boards, drive motors, limit switches, and so forth, to restore the antenna group to operation. Components and piece parts incident to, but not part of, modules or PC boards may be replaced by the technician.

3.7.3 Special tools. Special tools shall not be required for maintenance.

3.7.4 General purpose electronic test equipment (GPETE). GPETE shall be selected from MIL-STD-1364.

3.8 Identification and marking. The antenna group shall conform to the requirements specified in 3.8.1 through 3.8.6.

3.8.1 Antenna group identification. Marking of all antenna groups shall be in accordance with the requirements of MIL-P-15024.

3.8.2 Visibility. Parts shall be so mounted that their identification markings are readily visible. Auxiliary markings, such as on access panels, shall be provided so as to identify those parts where labeling is not readily visible.

3.8.3 Readability of identification plates. All identification (nomenclature) plates and serial numbers shall be located so they are readable after antenna group installation. They shall be located on the front surface of control panel of below deck equipment and on the side of above deck equipment.

3.8.4 Serial numbers. Serial numbers shall be assigned to all major assemblies, such as antenna control, deck box, antenna unit, and switching unit.

3.8.5 Identification of wiring. Wiring shall be identified by color-coding, or by other means, such as marking of terminals at both ends of each lead. Color-coding for chassis wiring shall be a system conforming to MIL-STD-681.

3.8.6 Markings. The reflector shall be stencilled with letters no less than two inches high "DO NOT PAINT WITH LEAD PAINT". Instructions shall be provided on an instruction plate for repainting, including the identification of paint type which will not degrade antenna group performance.

3.9 Antenna group power. The antenna group shall conform to nominal power requirements specified in a through c:

a. 440 volts, 3 phase, 3 wire (delta), 50 to 60 Hz, 2.5 kilovoltampere (KVA) (nominal), 0.92 power factor
b. 115 volts, 1 phase, 2 wire, 400 Hz (nominal), 250 voltampere (VA) (nominal), 1.0 power factor, from auxiliary reference voltage
c. 115 volts, 1 phase, 2 wire, 400 Hz (nominal), 250 VA (nominal), 1.0 power factor, from ship's gyro compass

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3.10 Workmanship screen. The antenna group shall withstand a defect detection screen of temperature cycling as specified in 4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilty for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to ensure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or unit conformance of product to prescribed criteria has been demonstrated.

4.1.2 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

1. Production inspection (see 4.4.1)
2. Production control inspection (see 4.4.2)
3. Environmental inspection (see 4.4.3)

c. Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.4), one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

TABLE II. First article inspection.

Examination or test	Requirement paragraph	Test paragraph
Surface examination	3.6, 3.7, 3.8	4.5.1
Operating test	3.1, 3.1.1.1, 3.1.1.2, 3.1.1.3, 3.1.1.4	4.5.2
Weights and dimensions	3.1.1.1.3, 3.1.1.3.7, 3.1.1.4.1.3	4.5.1
Supply line voltage and frequency and power	3.9	4.5.1
Enclosure	3.5.1.10	4.5.1
Temperature	3.5.1.1	4.5.1
Humidity	3.5.1.11	4.5.3
Sea States	3.5.1.6	4.5.1
Shock	3.5.1.7	4.5.1
Vibration	3.5.1.8	4.5.1

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TABLE II. First article inspection. (Continued)

Examination or test	Requirement paragraph	Test paragraph
Inclination	3.5.1.12	4.5.1
Salt fog	3.5.1.2	4.5.4
Altitude	3.5.1.3	4.5.7
Wind speed and icing	3.5.1.4, 3.5.2, 3.5.1.5	4.5.5
Nuclear air blast	3.5.1.15, 3.5.2	4.5.5
Nuclear hardening	3.5.1.13, 3.5.1.14	4.5.6
Radio frequency interference	3.5.1.9	4.5.1
Reliability	3.3	4.7

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every antenna group offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall include the requirements of TABLE III.

TABLE III. Production inspection.

Examination or test	Requirement paragraph	Test paragraph
Surface examinations	3.6, 3.7, 3.8	4.5.1
Operating test	3.1, 3.1.1.1, 3.1.1.2, 3.1.1.3, 3.1.1.4	4.5.2
Workmanship Screen	3.10	4.6

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to TABLE IV and to the inspection procedures of MIL-STD-105 using the special inspection level. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on antenna groups that have passed production inspection. The antenna groups shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

TABLE IV. Production control inspection.

Examination or test	Requirement paragraph	Test paragraph	AQL percentage
Weights and dimensions	3.1.1.1.3, 3.1.1.3.7, 3.1.1.4.1.3		6.5
Supply line voltage and frequency and power	3.9	4.5.1	6.5
Enclosure	3.5.1.10	4.5.1	6.5
Radio frequency interference	3.5.1.9	4.5.1	6.5
Reliability	3.3	4.7	6.5

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4.4.3 Environmental inspection. Environmental inspection shall be as listed in TABLE V and shall be performed on units which have been subjected to and passed production inspection.

TABLE V. Environmental inspection.

Examination or test	Requirement paragraph	Test paragraphs
Wind speed and icing	3.5.1.4, 3.5.2, 3.5.1.5	4.5.5
Nuclear air blast	3.5.1.15, 3.5.2	4.5.5
Nuclear hardening	3.5.1.13, 3.5.1.14	4.5.6
Temperature	3.5.1.1	4.5.1
Humidity	3.5.1.11	4.5.3
Shock	3.5.1.7	4.5.1
Vibration	3.5.1.8	4.5.1
Inclination	3.5.1.12	4.5.1
Salt fog	3.5.1.2	4.5.4
Altitude	3.5.1.3	4.5.7
Sea States	3.5.1.6	4.5.1

4.4.3.1 Environmental inspection samples. Samples for the environmental inspection shall be selected in accordance with the following:

a. One sample from the first 25 production units fabricated
b. One sample from each successive 100 units, or fraction thereof, produced

4.5 Test methods.

4.5.1 Test and examinations. Tests and examinations shall be conducted in accordance with the tests and examinations specified in MIL-E-16400 and 4.5.2 through 4.5.6.

4.5.2 Operating test. The antenna group shall be energized and subjected to an operating test to ensure the proper functioning and performance at an adequate number of points in the required range to demonstrate conformance to the requirements specified herein.

4.5.3 Humidity test. Humidity test shall be conducted in accordance with MIL-STD-810, Method 507, Procedure IV.

4.5.4 Salt fog test. Salt fog test shall be performed in accordance with MIL-STD-810, Method 509.

4.5.5 Wind speed, icing, and nuclear air blast. Testing shall be performed to demonstrate conformance to the requirements of 3.5.1.4, 3.5.1.5, and 3.5.1.15.

4.5.6 TREE and EMP testing. Tests shall be conducted on the preamplifier, power supply, and the resistive loads to demonstrate TREE hardening in compliance with 3.5.1.10. Testing of EMP hardening, however, is not mandatory for the antenna group (see 6.2). At the option of the Government, testing may be conducted at a Government test facility to demonstrate conformance with the EMP hardening of 3.5.1.9 (see 6.2).

4.5.7 Altitude. The antenna group shall be subjected to a low pressure test in accordance with MIL-STD-810, Method 500.1, Procedure I, except that operation altitude shall be 5000 feet (ft).

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4.6 Workmanship screen. All antenna groups shall withstand a defect detection vibration screen and temperature cycling screen as specified in 4.6.1 and 4.6.2.

4.6.1 Random vibration screen. Prior to conducting temperature cycling, vibration shall be performed on all antenna groups. The antenna groups shall be mounted in their normal operational position (without shock isolators) and the vibration stress applied vertical to this position for an accumulated time of 10 minutes. The vibration shall be in accordance with FIGURE 2. The control accelerometer shall be located next to one of the mounting points of the item under test. The acceleration spectral density (ASD) shall be controlled by an accelerometer signal with a filtered bandwidth of 10 Hz up to a vibration frequency of 500 Hz, and a 100 Hz bandwidth above 500 Hz. The test item shall be energized during vibration and appropriate input signal applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during test shall require corrective action before the test is resumed.

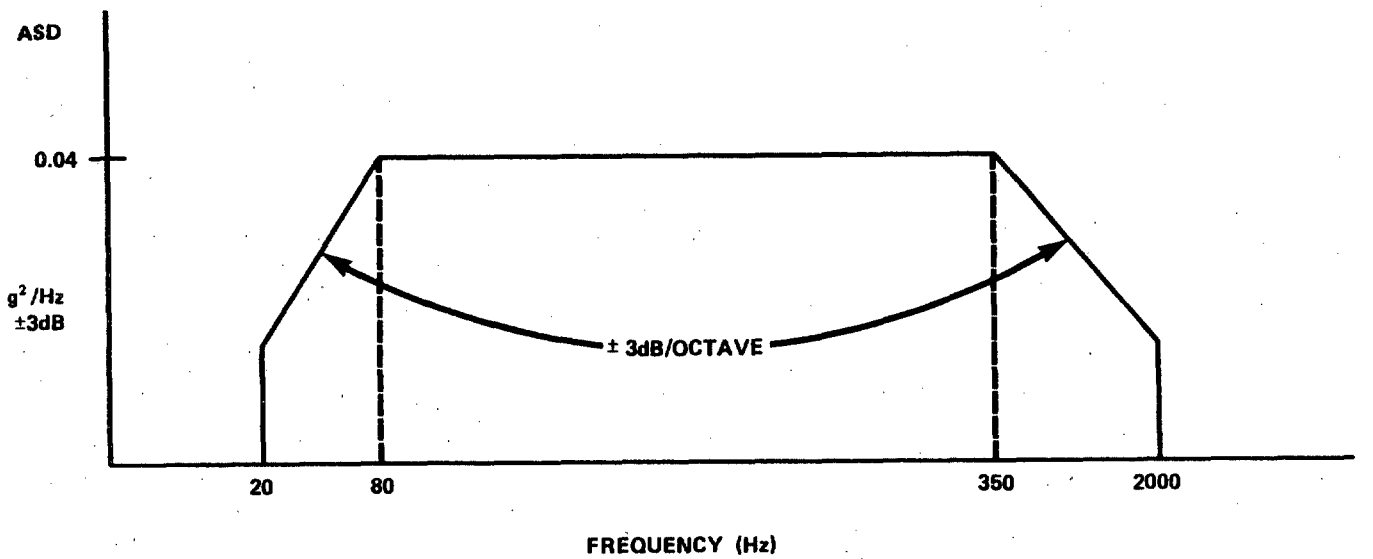


FIGURE 2. Random vibration.

4.6.2 Temperature screen. Each antenna group shall be subjected to 10 cycles of the temperature curve as specified in FIGURE 3. The temperature rate of change shall be not less than 5° C per minute. Antenna group power shall be turned on and off at the indicated times. The antenna group shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the antenna group reaches 80 percent of the chamber temperature. When failures occur, the antenna group shall be reworked and the cycling continued for a cumulative total of 10 cycles.

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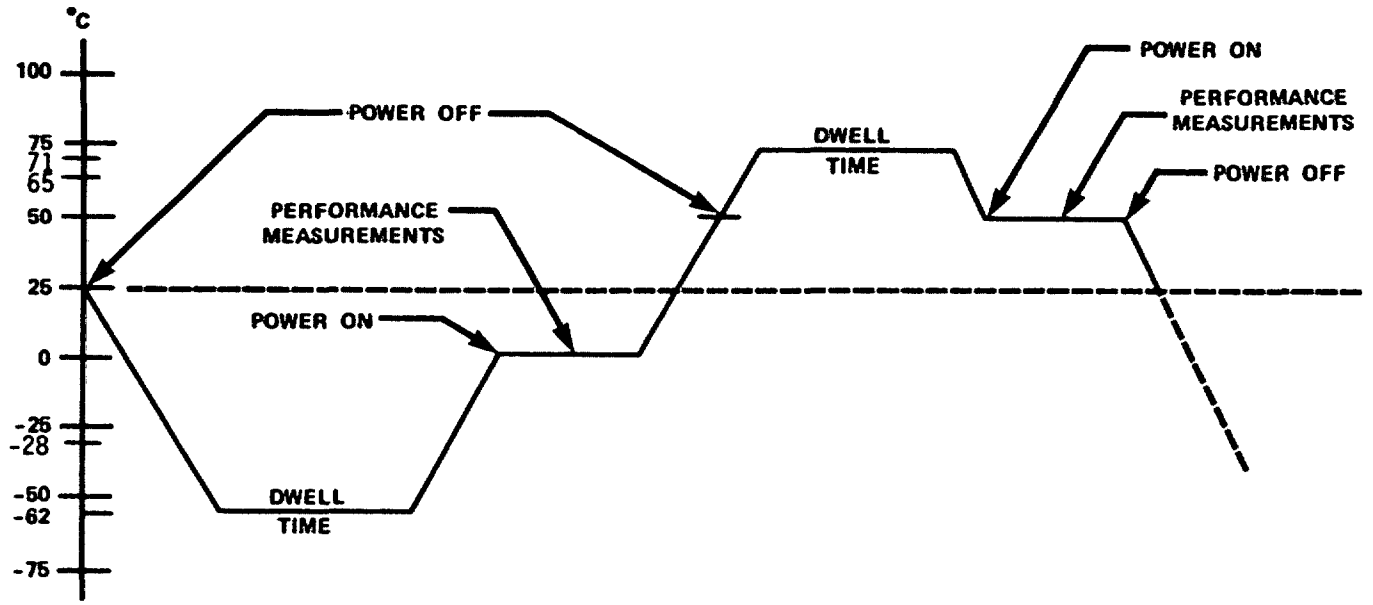


FIGURE 3. One cycle of temperature curve (not to scale).

4.7 Reliability.

4.7.1 Reliability demonstration. Reliability demonstration testing shall be performed on the first 4 antenna groups. Testing shall proceed to a decision in accordance with the criteria specified for category 3B equipment, test plan VIIIC of MIL-STD-781. This testing shall be completed to an accept decision before any production antenna groups can be accepted.

4.7.2 Relevant failure. The final decision as to the relevancy or non-relevancy of a failure shall be made by the command or agency concerned using the Failure categories criteria of MIL-STD-781 as a guide (see 6.5).

4.7.3 Rejected antenna groups. When the antenna group has been rejected as a result of a reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is antenna group manufacturer design or part manufacturer designs, corrective action shall be performed on all antenna groups delivered under the contract.

4.8 Inspection of preparation for delivery. Inspections shall be conducted to ensure compliance to the requirements of Section 5.

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5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, and packing specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The antenna group described herein is intended for shipboard use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples, if other than specified (see 4.3)
c. Levels of preservation, packaging, packing, and marking (see 5.1)
d. Whether EMP hardening is required for the antenna group (see 4.5.5)
e. If testing of EMP hardening is to occur at a Government facility (see 4.5.5)
f. Use of other than current documents listed in the DoDISS (see 2.1)

6.3 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified herein will be developed as specified by an approved Data Item Description (DID) (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified herein will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph	Data requirement	Applicable DID
4.5	Test procedures	UDI-T-22710
4.4	Quality conformance inspection report	UDI-R-22572

(Copies of DIDs required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the DAR. The first article should be a first production item. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

6.5 Definition.

Relevant failure. Any operating discrepancy or degradation that requires an unscheduled adjustment or calibration to be made, after initial satisfactory operation of the antenna group under test.

Preparing Activity:
NAVY-EC
(Project 5985-N495)

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APPENDIX

DERATING CRITERIA FOR ELECTRICAL AND ELECTRONIC PARTS

10. SCOPE

10.1 Scope. This appendix covers the derating criteria for electrical and electronic parts.

20. REQUIREMENTS

20.1 Unless otherwise approved by the Navy, these derating criteria apply to the fabrication during all acquisition phases.

20.2 Derating is increasing the ratio (margin of safety) between part design limits and the allowable applied stresses (electrical, thermal, mechanical, and other environmental stresses). Derating can be accomplished by either selecting a part with broader limits or by limiting the applied stresses. Derating provides added protection for part variances and decreases the part degradation rate thus increasing the expected life of a part.

20.3 The following criteria provide basic information for the derating of electrical and electronic parts and indicate the maximum allowable application stress values to be considered during console design. Since the safety margins of a part to both the threshold and time dependent stresses are generally unknown, parts should be derated to the maximum extent possible consistent with good design practices.

20.4 When derating, the designer shall consider the part manufacturer's specified environmental and operating condition rating versus the actual environmental and operating condition rating of the application and then apply derating factors so as not to exceed the maximum allowable application stresses herein.

Example: A wirewound resistor, RW67U--, which is rated 6.5 watts at 25°C, linearly derating to zero watts at 275°C, being operated in an equipment ambient of 70°C.

a. Compute rated power at 70°C;

$$P_{70}\ \text{rated} = P_{25}\ \text{rated} \times \frac{T_{max} - T_{operating}}{T_{max} - T_{rating}}$$

where $T_{operating} \times T_{rating}$

$$= 6.5 \times \frac{275 - 70}{275 - 25} = 6.5 \times \frac{205}{250}$$

$$= 5.33 \text{ watts}$$

b. Apply derating criteria: from tables - limiting power to 0.50 Max P = 5.33 x 50 percent = 2.66 watts

PART TYPE	CRITICAL STRESS	MAXIMUM ALLOWABLE APPLICATION STRESS	REMARKS
MICROCIRCUITS			
Digital	Output current	0.80	
	operating frequency	0.75	
Linear	Bias voltage	0.70	
	Input signal voltage	0.70	

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PART TYPE	CRITICAL STRESS	MAXIMUM ALLOWABLE APPLICATION STRESS	REMARKS
	Output current	0.75	
	operating frequency	0.75	
Voltage	Input voltage (rated maximum)	0.80	
	Output current (rated maximum)	0.75	
	Power dissipation (rated maximum)	0.60	
All	Junction temperature	110°c MAX	
All	Ambient temperature	85°c MAX	
TRANSISTORS	See 1/		
General purpose	Power	0.50	
	Current	0.75	
	Voltage	0.75	
Power	Power	0.30	
	Current	0.75	
	Voltage	0.75	
Switching	Power	0.50	
	Current	0.75	
	Voltage	0.75	
All	Junction temp	110°C MAX	
	Leakage current	See remarks	Consider Leakage current to be at least twice rated value.
	Frequency	2/	
DIODES			
General purpose	Power	0.50	
	Peak inverse voltage (PIV)	0.50	
	Surge current	0.50	
	FWD current	0.50	
Rectifier	Power	0.30	
	PIV	0.50	
	Surge current	0.50	
	FWD Current	0.50	
Switching	Power	0.30	
	PIV	0.50	
	Surge current	0.50	
	FWD current	0.50	

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PART TYPE	CRITICAL STRESS	MAXIMUM ALLOWABLE APPLICATION STRESS	REMARKS
SCR	Power	0.30	
	PIV	0.50	
	Surge current	0.50	
	FWD current	0.50	
Varactor	Power	0.50	
	PIV	0.75	
	FWD current	0.75	
Zener	Power	0.50	
	FWD current	0.50	
	Zener current	0.5 $I_{z_{max}}$ + $I_{z_{nom}}$	
Reference	Power	0.30	
	FWD current	0.50	
	Zener current	0.5 $I_{z_{max}}$ + $I_{z_{nom}}$	
All	Junction temp	110°C MAX	

PART TYPE	CRITICAL STRESS	MAXIMUM ALLOWABLE APPLICATION STRESS 3/	REMARKS
CAPACITORS, FIXED	SEE 4/		
Ceramic	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Mica, dipped	Voltage	0.60	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Glass	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Porcelain	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Paper	Voltage	0.50	Good low frequency characteristics
	Current		
	AC/ripple	0.70	
	Surge	0.70	

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PART TYPE	CRITICAL STRESS	MAXIMUM ALLOWABLE APPLICATION STRESS 3/	REMARKS
Plastic	Voltage	0.50	Mylar good to 1 MHz; teflon, polystyrene, and polycarbonate good to 1 GHz
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Tantalum, foil or film	Voltage	0.50	Limit to 70°C ambient temperature
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Tantalum, solid, dry	Voltage	0.50	Limit to 85°C ambient temperature
	Current		
	AC/ripple	0.70	
	Surge protection	3 ohms/volt	
Tantalum, solid, wet	Voltage	See remarks	Use of tantalum wet slug capacitors is prohibited except by prior approval of the procuring activity. 5/
	Current		
	AC/ripple	See remarks	
CAPACITORS ADJUSTABLE			
Ceramic	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Glass	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Air	Voltage	0.30	
All, fixed & adjustable	Leakage	See remarks	For all applications, consider leakage current as twice rated maximum value

PART TYPE	CRITICAL STRESS	MAXIMUM ALLOWABLE APPLICATION STRESS	REMARKS
RESISTORS, FIXED	See 6/		
Carbon composition	Power	0.50	
Insulated film	Power	0.50	

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PART TYPE	CRITICAL STRESS	MAXIMUM ALLOWABLE APPLICATION STRESS	REMARKS
Wirewound precision			
1.0 percent total	Power	0.50	
0.1 percent total	Power	0.25	
Wirewound power	Power	0.50	
Thermistor	Power	0.50	Uses in other than zero power applications shall specify minimum wattage for the application.
ADJUSTABLE	SEE 6/		
Wirewound	Rated current	0.70	See 7/
Non-Wirewound	Rated current	0.70	See 7/
All	Voltage	0.80	Maximum voltage shall not exceed 80 percent of the MIL-Ratings.
RELAYS AND SWITCHES LOAD TYPE			
Capacitive	Contact current	0.75	Peak inrush current shall not exceed derated limit.
Resistive	Contact current	0.75	
Inductive	Contact current	0.40	
Motor	Contact current	0.20	
Filament	Contact current	0.10	
MISCELLANEOUS			
Transformer and chokes	Flux saturation level	0.75	Parts shall be used at winding temperature at least 30°C below rating. Maximum voltage between windings or between winding and shield shall not exceed 60 percent of rating for steady-state conditions or 90 percent of rating with transients.
	Insulation breakdown voltage	0.50	
	Current	0.70	

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PART TYPE	CRITICAL STRESS	MAXIMUM ALLOWABLE APPLICATION STRESS	REMARKS
Connectors	Current, pin	See remarks	Current derating for connector pins shall be in accordance with criteria for relays and switches, depending on type of loading. For paralleled pins sum of current must not be more than 100 percent rated of lowest rated pin.
	Insulation breakdown voltage	0.50	

1/ Worst-case combination of DC, AC, and transient voltages shall be no greater than derated limit.

2/ Transistors rated at 500 milliwatts or less in TO-5 or TO-18 packages using one MIL or smaller aluminum lead wires with thermocompression bonds shall not be used at repetition rates of less than 30 Hz. At 30 Hz and 25°C or less, derate the power dissipation to 0.20 and limit the input current to 50 mA. Above 25°C, derate an additional 1.33 mW/°C. Gold lead devices should be used at repetition rates less than 30 Hz.

3/ Manufacturer's derating factors shall be applied before applying these factors.

4/ Voltage equals instantaneous total of DC, AC, surge, and transient voltages.

5/ When approved for use, tantalum wet slug capacitors shall be subjected to special screens such as temperature cycling and seal leakage.

6/ High density packaging may require further derating if ambient temperatures are increased.

7/ Rated current is defined as:

$$I_R = \sqrt{\frac{P_{max}}{R_{max}}}$$

And by limiting the current to 0.70 rated current, power is limited to 0.50 maximum power.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-A-28865(EC) Antenna Group, OE-82( )/WSC-1(V)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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